

Foundation Models: Modern Advances and Applications

Naeemullah Khan

naeemullah.khan@kaust.edu.sa



جامعة الملك عبد الله
للعلوم والتقنية
King Abdullah University of
Science and Technology

KAUST Academy
King Abdullah University of Science and Technology

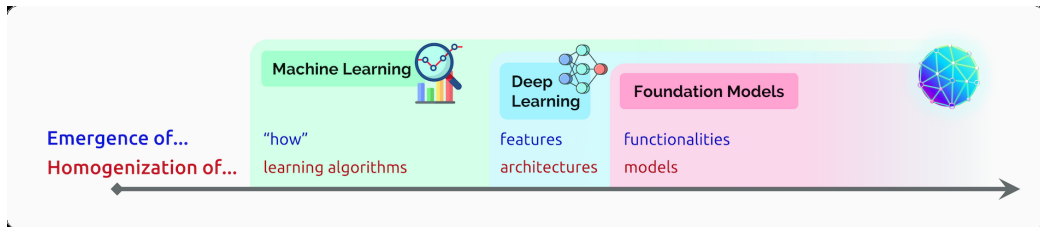
July 6, 2026

1. Recap: from self-supervised pretraining to foundation models
2. Foundation models: definition, impact, types, and adaptation (probes, LoRA, prompting)
3. Foundation models vs. LLMs and ecosystem implications (Bommasani et al.)
4. Historical arc and vision FM taxonomy (representation vs. generative)
5. CLIP: joint image–text representations and zero-shot transfer
6. Grounding DINO and DINOv2: open-vocabulary detection and representation learning
7. SAM 1, SAM 2, and SAM 3: promptable segmentation and data engines
8. Depth Anything and SuperGlue: monocular depth and geometric matching
9. TabPFN and TRIBE v2: tabular and brain-response foundation models
10. InternVideo2: large-scale video foundation modeling
11. Limitations, summary, hands-on resources, and references

- ▶ **SSL and contrastive lectures:** pretext tasks, contrastive learning (SimCLR, MoCo), and self-distillation (BYOL, DINO, DINOv2) learn features from unlabeled data.
- ▶ **JEPA lecture:** latent prediction (I-JEPA, V-JEPA 2) trains encoders that serve many tasks from frozen features; V-JEPA 2 already behaves like a video foundation model.
- ▶ Those lectures answered *how to pretrain*. Today we look at the pretrained models the field actually builds on: what they are, what data they use, and how they are adapted.
- ▶ Working definition (Bommasani et al., 2021): a **foundation model** is trained on broad data at scale and can be **adapted** to a wide range of downstream tasks.

What are Foundation Models?

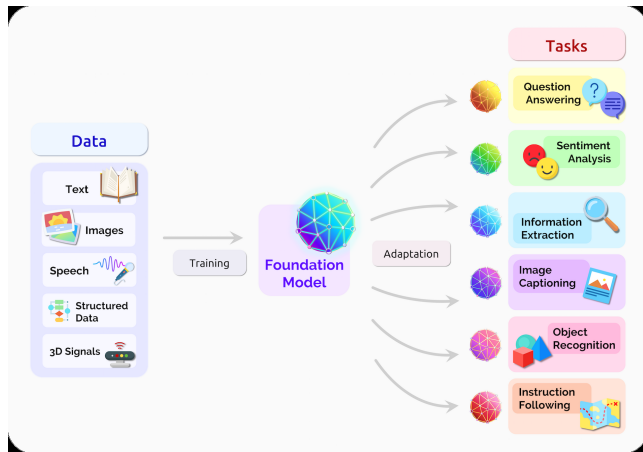
- ▶ **Foundation models** are large, general AI models pretrained on massive, diverse datasets.
- ▶ They provide a universal base that can be adapted (via fine-tuning or prompting) across many domains and tasks.
- ▶ The key shift: build powerful, flexible models for reuse, rather than training a new model for each task from scratch.



Emergence and homogenization from ML → deep learning → foundation models (Bommasani et al., arXiv:2108.07258).

- ▶ **Accelerate innovation:** Dramatically reduce time and resources needed to create high-performing AI applications.
- ▶ **Expand accessibility:** Lower the expertise required for deploying AI, empowering domain experts (not just AI researchers).
- ▶ **Enable new capabilities:** Support advanced “few-shot” and “zero-shot” learning, allowing for rapid adaptation to new tasks.
- ▶ **Transform industries:** Fuel advances in medicine, science, robotics, business, and beyond.
- ▶ **Introduce new challenges:** Require careful management of data/computational resources, fairness, safety, and responsible use.

One Foundation Model, Many Tasks



Diverse modalities feed training; one foundation model supports many downstream tasks via adaptation (Bommasani et al., arXiv:2108.07258).

- ▶ **Language models:** Trained on text data, these models can generate, summarize, and understand language.
Examples: BERT, GPT
- ▶ **Vision models:** Trained on images/videos, they can recognize objects, segment scenes, and more.
Examples: CLIP, SAM
- ▶ **Multi-modal models:** Learn to connect information across text, images, audio, or video.
Examples: CLIP (text-image), Flamingo, GPT-4V, Gemini
- ▶ **Other domains:** Foundation models exist for structured (tabular) data, medical data, code, and scientific domains.
Examples: TabPFN (tabular), TRIBE (brain responses); both appear later in this lecture

- ▶ **Pretraining at scale:** Train on enormous datasets (text, images, etc.) using self-supervision; learn generalized representations.
- ▶ **Adaptation:** Foundation models can be quickly adapted to new tasks using:
 - Fine-tuning on small amounts of labeled data
 - Prompting (providing instructions or context)
 - In-context learning (few-shot or zero-shot)
- ▶ **Transfer learning:** Reuse knowledge across tasks and domains, reducing the need for bespoke model training.
- ▶ **Unified architectures:** Many foundation models use architectures like transformers, enabling processing across modalities.

- ▶ **Frozen probes:** keep the backbone fixed and train a small head on top (linear or attentive probe, as in the JEPA lecture). Cheapest option; also the standard way to *evaluate* a foundation model.
- ▶ **Full fine-tuning:** update all weights on the downstream task. Strongest task performance, highest cost, and the model can forget general capabilities.
- ▶ **Parameter-efficient fine-tuning (PEFT):** train small added modules while the backbone stays frozen. **LoRA** (Hu et al., 2022) learns a low-rank update $\Delta W = BA$ next to each frozen weight matrix W , typically well under 1% of the parameters.
- ▶ **Prompting and in-context learning:** no weight updates at all; instructions or examples are given at inference time. TabPFN (later this lecture) classifies whole tabular datasets this way.

Foundation models

- ▶ Trained on **heterogeneous** data: text, images, speech, structured data, 3D/signals, and more.
- ▶ **Homogenization**: one pretrained model is adapted to many applications; **emergence**: capabilities (e.g. in-context learning) from scale were not explicitly programmed.
- ▶ Objectives: self-supervised, contrastive, masked, or alignment losses, not only next-token prediction.

LLMs

- ▶ **Language-first**: data are primarily text; core loss is usually autoregressive or masked LM.
- ▶ **Downstream**: generation, QA, summarization, code; multimodal **variants** pair an LLM with other encoders.
- ▶ Same “pretrain then adapt” pattern, but the **scope of pretraining data** is narrower than full multi-modal foundation models.

- ▶ **Foundation models** enable transfer across domains (vision–language, robotics, biomedicine, ...) but concentrate capability in a few shared pretrained artifacts.
- ▶ **LLMs** excel at language and code; for vision, video, or tabular data they are usually **one component** of a larger system unless trained multi-modally from the start.
- ▶ **Limitations:** compute and data scale, inherited biases from pretraining, evaluation and safety gaps as the same “foundation” is reused widely.

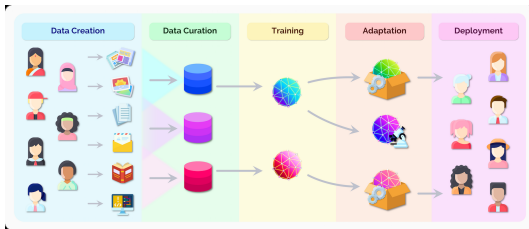
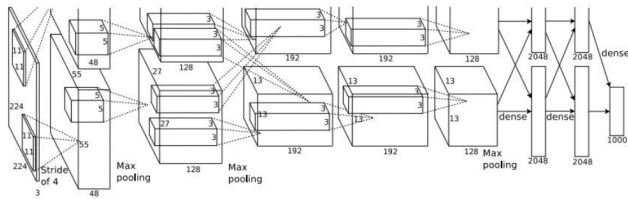
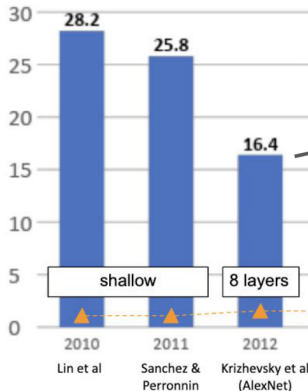


Fig. 3: From people and data through training and adaptation to deployment (Bommasani et al., arXiv:2108.07258).

2012: Deep learning emerges in AI

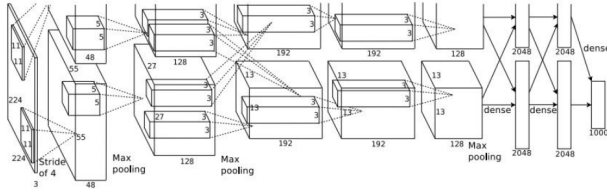


Krizhevsky et al. 2012. 8-layer "AlexNet".

ImageNet Visual Recognition Challenge results.

Convergence of key ingredients of deep learning

Algorithms



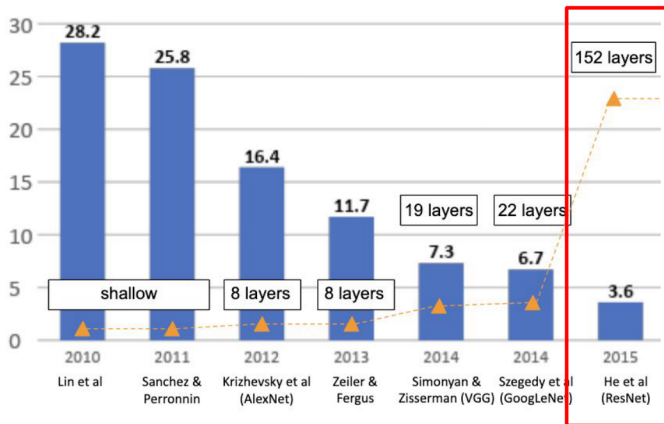
Compute



Data



2015: Very deep convnets and challenging vision tasks

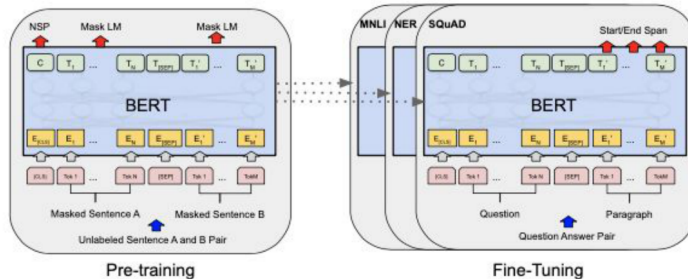


He et al. 2015. ResNet.



2018: Breakthroughs in deep learning for natural language processing (sequences)

Transformer neural network architectures (containing self-attention mechanism) and pre-training -> fine-tuning. State-of-the-art on 11 NLP benchmarks.



Vaswani et al. 2017. "Attention is all you need."
Devlin et al. 2018. BERT.

SYSTEM PROMPT (HUMAN-WRITTEN)

In a shocking finding, scientist discovered a herd of unicorns living in a remote, previously unexplored valley, in the Andes Mountains. Even more surprising to the researchers was the fact that the unicorns spoke perfect English.

MODEL COMPLETION (MACHINE-WRITTEN, 10 TRIES)

The scientist named the population, after their distinctive horn, Ovid's Unicorn. These four-horned, silver-white unicorns were previously unknown to science.

Now, after almost two centuries, the mystery of what sparked this odd phenomenon is finally solved.

Dr. Jorge Pérez, an evolutionary biologist from the University of La Paz, and several companions, were exploring the Andes Mountains when they found a small valley, with no other animals or humans. Pérez noticed that the valley had what appeared to be a natural fountain, surrounded by two peaks of rock and silver snow.

Transformer
architectures become
ubiquitous

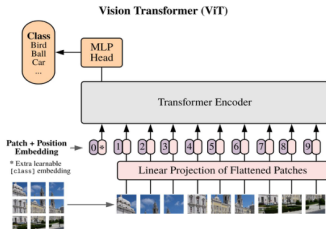
Training these models
on very large amounts
of data -> rise of
“foundation models”

OpenAI GPT-3 (figure from GPT-2). Brown et al. 2020.

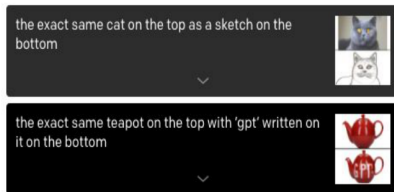
Transformer architecture can be applied to images as well!

Key idea: Convert image into sequence of patches (i.e. image tokens).

Can then benefit from Transformer architecture and its self-attention mechanism. Allows process of token encoding to incorporate contextual understanding (attention) from other tokens of the same (self) sequence.

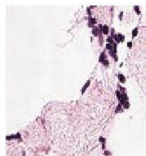


ViT. Dosovitsky et al. 2021



DALL-E. Ramesh et al. 2021.

healthy lymph node tissue (22.8%) Ranked 2 out of 2



✗ this is a photo of lymph node tumor tissue

✓ this is a photo of healthy lymph node tissue

CLIP. Radford et al. 2021.

2022: Major improvements in text-to-image generation models

DALL-E 2
Ramesh et al. 2022.



vibrant portrait painting of Salvador Dalí with a robotic half face

a shiba inu wearing a beret and black turtleneck

Meta Make-a-Video.
Singer et al. 2022

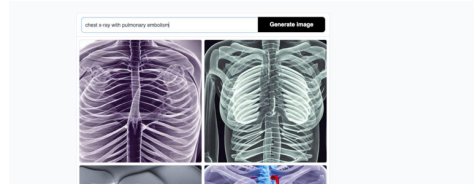


Stable Diffusion Online

Stable Diffusion is a latent text-to-image diffusion model capable of generating photo-realistic images given any text input, cultivates autonomous freedom to produce incredible imagery, empowers billions of people to create stunning art within seconds.

Create beautiful art using stable diffusion ONLINE for free.

Get started →



Stable Diffusion.
<https://stablediffusionweb.com/>

Late 2022: ChatGPT reaches public mainstream

User help me write a short note to introduce myself to my neighbor

ChatGPT Hi there,

My name is [Your Name] and I just moved in next door. I wanted to reach out and introduce myself. I'm looking forward to getting to know you and the neighborhood. If you ever need anything or just want to chat, please don't hesitate to stop by or give me a call.

Best,
[Your Name]

User can you make it more formal?

ChatGPT Dear Neighbor,

I am writing to introduce myself. My name is [Your Name] and I have recently moved in next door. I wanted to take a moment to reach out and say hello. I am looking forward to getting to know you and the neighborhood.

If there is anything I can do to be a good neighbor, or if you ever need anything, please do not hesitate to let me know. I am happy to help in any way that I can.

Sincerely,
[Your Name]

OpenAI ChatGPT

<https://openai.com/blog/chatgpt>

Representation Learners

Train on huge amounts of data

Objective: learn powerful generalist feature extractors that can be used for downstream computer vision tasks.

- ▶ Vision Only (e.g. DINOv2)
- ▶ Vision-Language (e.g. CLIP)

Generative Models

Train on huge amounts of data

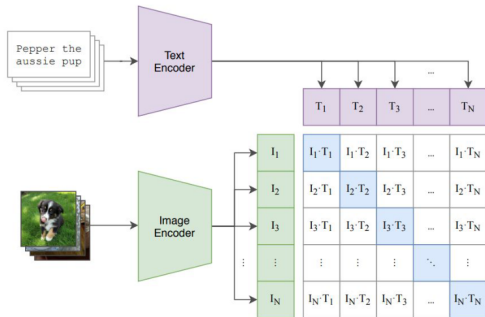
Objective: learn to generate images and/or text.

- ▶ Text $\rightarrow$ Image (e.g. Stable Diffusion, DALL-E)
- ▶ Text, Image $\rightarrow$ Text (e.g. Flamingo, GPT-4V)

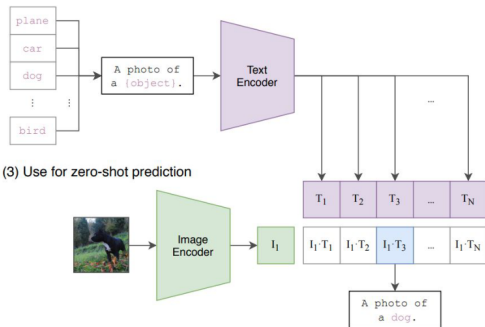
- ▶ **Text-to-image diffusion** (Stable Diffusion, DALL-E): start from random noise and iteratively **denoise** it, conditioned on a text prompt. The denoiser is trained on billions of captioned images; latent diffusion (Rombach et al., 2022) runs the process in a compressed latent space to cut cost.
- ▶ **Vision-language generative models** (Flamingo, GPT-4V, LLaVA): a vision encoder feeds visual tokens into an **LLM** through a small projector, the same pattern as the V-JEPA 2 language alignment seen last lecture. The LLM then generates text about the image.
- ▶ Text conditioning in both cases relies on strong text encoders; Stable Diffusion uses **CLIP's** text encoder, which we cover next.
- ▶ This lecture focuses on the **representation-learning** class; generative modeling has its own dedicated lectures in the full course.

- ▶ **CLIP** (Radford et al., 2021): trained on **400 million** image–text pairs collected from the web; no class labels at all.
- ▶ Two encoders: an **image encoder** (ResNet or ViT) and a **text encoder** (transformer), each mapping its input to a shared embedding space.
- ▶ Training pulls each image toward *its own caption* and away from the other captions in the batch.
- ▶ Same contrastive idea as SimCLR from the contrastive-methods lecture, but the two “views” come from **different modalities**: an image and its natural-language description.

(1) Contrastive pre-training



(2) Create dataset classifier from label text



(3) Use for zero-shot prediction

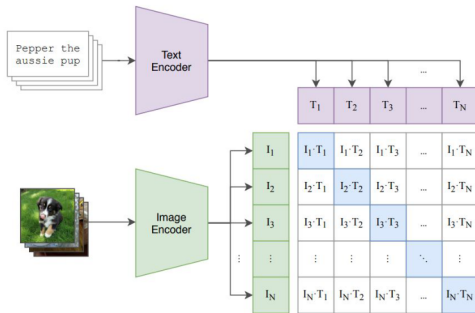
Radford et al. 2021.

For a batch of N pairs, compute the $N \times N$ matrix of cosine similarities s_{ij} between image embeddings I_i and text embeddings T_j . The loss is a symmetric cross-entropy that treats the diagonal as correct:

$$\mathcal{L} = -\frac{1}{2N} \sum_{i=1}^N \left[\log \frac{e^{s_{ii}/\tau}}{\sum_j e^{s_{ij}/\tau}} + \log \frac{e^{s_{ii}/\tau}}{\sum_j e^{s_{ji}/\tau}} \right]$$

- ▶ First term: match each **image** to the right **caption**; second term: match each **caption** to the right **image**.
- ▶ τ is a learned **temperature**; large batches (up to 32,768 pairs) supply many negatives, as in the contrastive methods seen earlier.
- ▶ The result is a **joint embedding space** where semantically matching images and texts are close.

(1) Contrastive pre-training



Radford et al. 2021.

Learn text and image encoders (i.e. feature extractors) through a self-supervised “contrastive” objective: images and their matching text captions should produce more similar features than non-matching ones

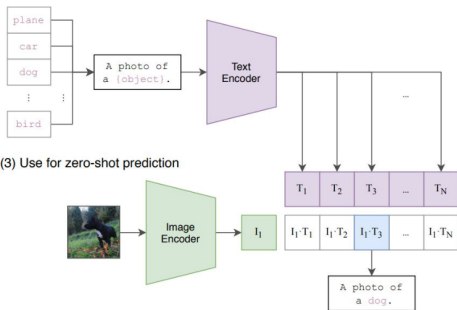
Trained on 400M image-text pairs: generated by searching for image-text pairs on the web, where text comes from a base query list of 500,000 queries comprising all words occurring at least 100 times in the English version of Wikipedia

- ▶ A classifier can be built **from label names alone**: embed prompts such as “a photo of a {label}” with the text encoder, embed the image, and pick the label with the highest similarity.
- ▶ No training on the target dataset: this is **zero-shot transfer**. Prompt wording matters, and ensembling several templates improves accuracy.
- ▶ The largest CLIP matches the accuracy of the original supervised ResNet-50 on ImageNet **without seeing a single labeled ImageNet example**, and is notably more robust to distribution shift.
- ▶ Known weaknesses: fine-grained categories, counting objects, and reading text in images; the web training data also carries social biases.

Due to the contrastive training objective, image and text encoders output features in a shared representation space and the trained CLIP model can be used for **zero-shot classification**, i.e., perform N-way classification without showing the model paired examples of (input, class) for any of the N classes.

Radford et al. 2021.

(2) Create dataset classifier from label text



- ▶ CLIP's encoders became **building blocks** across the field:
 - **Stable Diffusion** conditions image generation on CLIP text embeddings.
 - **Open-vocabulary detection** (Grounding DINO, next section) grounds detections in text.
 - **SAM 3** segments “any concept” described by a short text prompt.
 - Vision-language assistants (LLaVA and similar) use a CLIP-style vision tower.
- ▶ **OpenCLIP** and the **LAION** datasets reproduce CLIP openly, so the recipe is fully public.
- ▶ In practice: when a task can be phrased in language, a CLIP-family model often gives a strong zero-shot baseline before any training.

- ▶ Open-vocabulary detection with natural-language grounding.
- ▶ Uses transformer-based backbone for both image and text encoding.
- ▶ Matches object proposals in an image with user-provided text prompts (e.g. “a red backpack”).
- ▶ Produces bounding boxes, labels, and confidence scores for detected objects.
- ▶ **Key architectural mechanisms:**
 - **Feature enhancer neck** with deep image–text cross-attention (not only late fusion).
 - **Language-guided query selection** to initialize decoder queries on relevant regions.
 - **Cross-modality decoder head** for iterative region-text refinement.
- ▶ **Result:** strong transfer to unseen categories and prompts; the text encoder builds on the CLIP-style grounding idea from the previous section.

Grounding DINO: How It Works

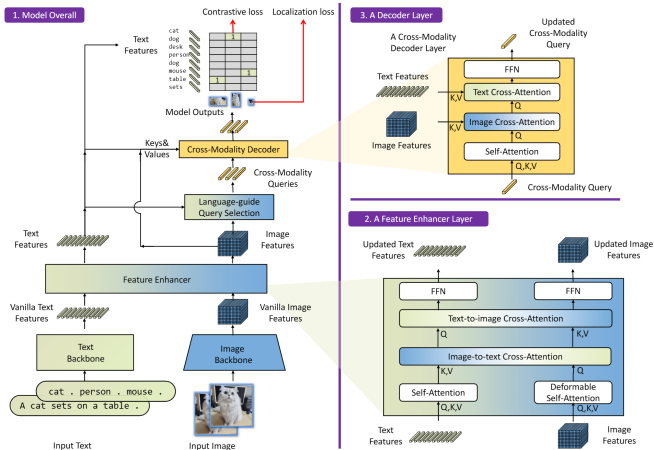
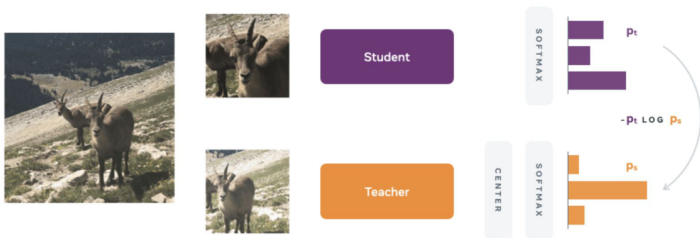


Figure 3. The framework of Grounding DINO. We present the overall framework, a feature enhancer layer, and a decoder layer in block 1, block 2, and block 3, respectively.

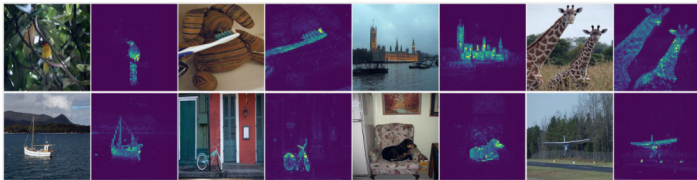
- ▶ Scales the **self-distillation** recipe from the contrastive-methods lecture (DINO image-level loss + iBOT masked-token loss) into a foundation model.
- ▶ **Curated data instead of raw web data:** LVD-142M is built by retrieving images similar to curated seed datasets from a large web pool, with deduplication. Data curation is a core part of the recipe.
- ▶ Train a large ViT (~ 1.1 B parameters), then **distill** it into smaller ViTs for deployment.
- ▶ **Frozen features** transfer to classification, segmentation, depth, and retrieval without fine-tuning; Depth Anything (later this lecture) uses a DINOv2 encoder.
- ▶ **DINOv3** (2025) scales further (7B-parameter ViT) and adds a gram-anchoring loss to keep **dense** patch features clean at long training schedules.

- DINO (self-Distillation with NO labels) is trained through self-supervised learning (no external label curation needed)
- Student-teacher self-supervised training objective: At a high level, images (teacher) should produce similar feature representations to randomly cropped portions of the image (student)



Caron et al. 2021, Oquab et al. 2023

- Once trained, the DINO neural network backbone can be used as the “pre-trained” feature extractor for many downstream tasks

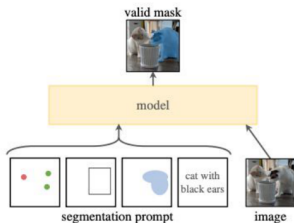


- DINOv2 made further implementation refinements and trained on 142 million images
- An example of a vision “**foundation model**”: a model trained on a very large amount of data that can be subsequently used or tuned for diverse tasks. DINOv2 was demonstrated to be a state-of-the-art backbone for many tasks.

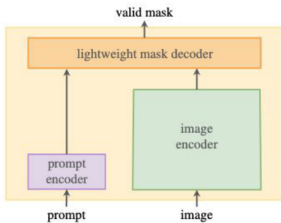
Caron et al. 2021, Oquab et al. 2023

Segment Anything Model (SAM): Promptable Segmentation

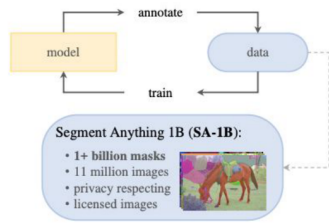
- Foundation Model for promptable segmentation, based on a Transformer encoder-decoder architecture
- Trained on 1 billion masks from 11 million images, using the model in a data collection loop



(a) **Task:** promptable segmentation



(b) **Model:** Segment Anything Model (SAM)



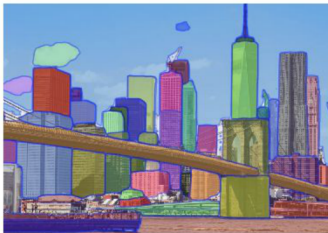
(c) **Data:** data engine (top) & dataset (bottom)

Kirrilov et al. 2023.

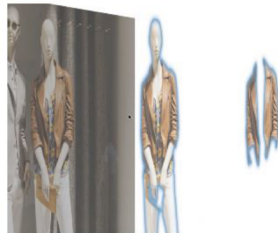
- Allows segmentation based only from prompts, without needing to provide training examples of target classes (zero-shot segmentation)



Prompt it with interactive points and boxes.



Automatically segment everything in an image.

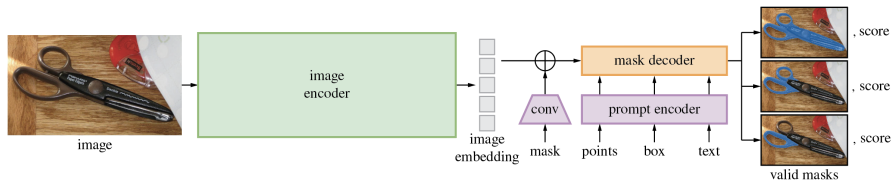


Generate multiple valid masks for ambiguous prompts.

Meta AI Segment Anything Model (SAM)
<https://segment-anything.com/>

Kirrilov et al. 2023.

- ▶ A heavy **image encoder** (MAE-pretrained ViT) embeds the image once; a light **prompt encoder** and **mask decoder** then run interactively in milliseconds.
- ▶ User or downstream model provides prompts (points, boxes, masks).
- ▶ The decoder predicts **several candidate masks** to resolve ambiguous prompts (a click on a shirt could mean the shirt or the person).
- ▶ Produces high-quality object masks across a wide range of settings without fine-tuning; masks are class-agnostic (no labels attached).



- ▶ There was no billion-mask dataset to train on, so Meta built one **with the model in the loop**, in three stages:
 1. **Assisted-manual:** annotators refine masks proposed by an early SAM.
 2. **Semi-automatic:** SAM proposes confident masks; annotators add what it missed.
 3. **Fully automatic:** SAM is prompted with a grid of points and annotates images alone.
- ▶ Result: **SA-1B** with 11M licensed images and **1.1B masks**, orders of magnitude larger than prior segmentation datasets.
- ▶ The pattern (model annotates, humans verify, retrain, repeat) recurs in SAM 2 and SAM 3; for foundation models, the **data engine** is as important as the architecture.

- ▶ Extends promptable segmentation to **video**: prompt on any frame, and the model propagates the object mask (a *masklet*) through the clip.
- ▶ **Streaming memory**: a memory bank stores features and object pointers from past frames; each new frame **cross-attends** to this memory before decoding its mask (the attention mechanism from the transformer lectures).
- ▶ Corrections are interactive: extra clicks on any frame refine the whole track.
- ▶ Trained with its own data engine, producing the **SA-V** dataset ($\sim 51\text{K}$ videos, $\sim 600\text{K}$ masklets); also faster than SAM 1 on still images.

SAM 1

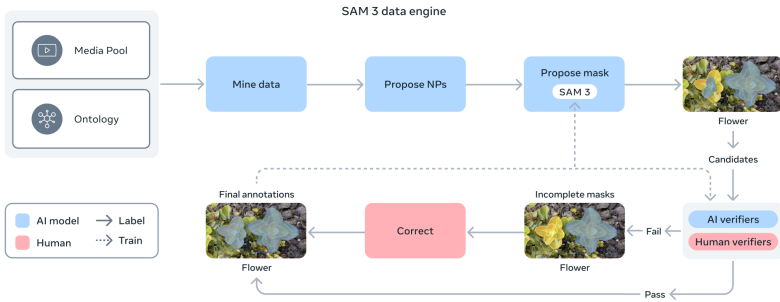
- ▶ Segments any object in any image.
- ▶ Requires as little as a single click to segment an object.
- ▶ Refine prediction with a few clicks.

SAM 2

- ▶ Segments and tracks any object in any image or video.
- ▶ Uses click, box, or mask prompts.
- ▶ Tracks segment predictions across video frames.
- ▶ Refine prediction with a few clicks.

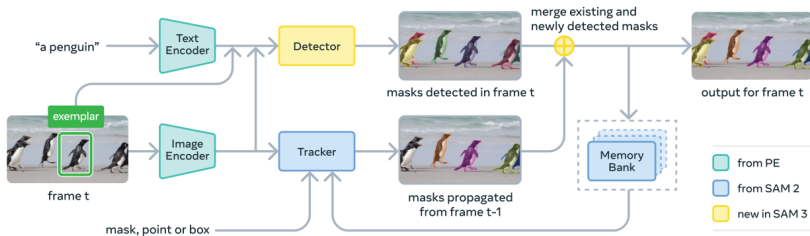
SAM 3

- ▶ Detect, segment, and track every instance of any object category in image or video.
- ▶ Uses text or examples as prompts.
- ▶ Segments and tracks objects from a click.
- ▶ Tracks segment predictions across frames.
- ▶ Detects and segments matching instances.
- ▶ Refine detection with visual examples.



SAM 3 data-engine loop from Meta blog post.

- ▶ Getting masks *with text labels* across many categories and domains is the hard part; SAM 3 scales the SAM data-engine pattern to concepts.
- ▶ AI models and annotators (human and Llama-based) collaborate in a feedback loop to mine, annotate, and verify data, yielding **4M+** annotated concepts across diverse domains.



SAM 3 model summary and detector/tracker architecture.

A dual encoder-decoder transformer serves as the **detector** for image-level capabilities and is used together with a **tracker** and memory for video. Both receive vision-language features from an aligned Perception Encoder (PE) backbone.

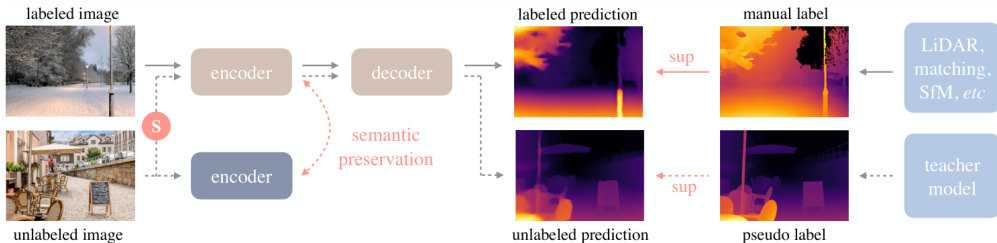


SAM 3.1 multiplexed inference versus per-object computation.

- ▶ Process multiple tracked objects in one forward pass (*multiplexing*) instead of one pass per object.
- ▶ Shared object-level context reduces redundant compute and helps crowded-scene tracking.

- ▶ Foundation model for single image depth estimation.
- ▶ Trained with a large-scale teacher–student pipeline:
 - $\sim 1.5\text{M}$ labeled images + $\sim 62\text{M}$ unlabeled images
- ▶ Encoder priors from DINOv2 improve cross-domain geometric understanding.
- ▶ Robust transfer across indoor/outdoor domains with limited task-specific supervision.
- ▶ Depth Anything V2 (2024) trains the teacher on synthetic images with exact labels, improving fine detail and robustness.
- ▶ Applications:
 - Robotics and autonomous driving
 - Virtual/augmented reality
 - 3D scene understanding and reconstruction

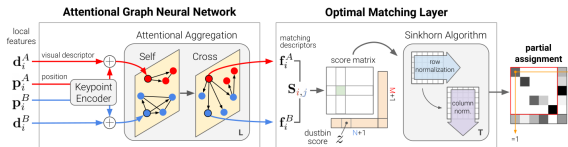
- ▶ **Teacher–student pipeline** on labeled depth plus large-scale unlabeled RGB: pseudo-labels from the teacher.
- ▶ DINOv2-style encoder + prediction head maps RGB to per-pixel depth.
- ▶ Outputs dense depth maps that transfer to segmentation and other geometric downstream tasks.



Yang et al., “Depth Anything,” CVPR 2024 ([OpenAccess](#)).

- ▶ Combines graph neural networks and attention to find matches between sets of keypoints in different images.
- ▶ Generalizes across scenes and keypoint detector types.
- ▶ Applications: Visual localization and mapping, 3D reconstruction, Object pose estimation.
- ▶ A caveat on terminology: SuperGlue (2020) predates the foundation-model era and is task-specific; it appears here as the standard *reusable component* for matching, a role foundation models now play for other tasks.

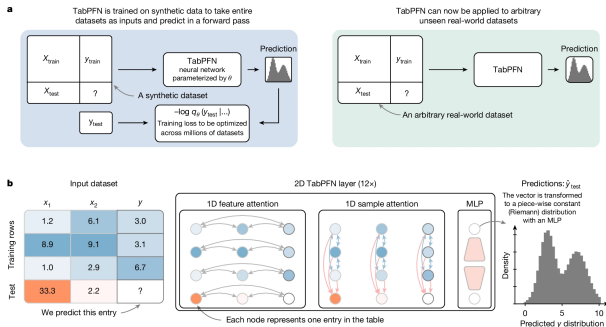
- ▶ Two main components: an attentional graph neural network (GNN) and an optimal matching layer.
- ▶ GNN uses keypoint encoder to combine each keypoint's position and visual descriptor into single vector.
- ▶ Alternating self-attention and cross-attention layers create powerful feature representations.
- ▶ The optimal matching layer builds an $M \times N$ score matrix between keypoints of both images, appends additional rows/columns ("dustbins") for potential unmatched keypoints.
- ▶ Uses Sinkhorn algorithm for T iterations to find optimal partial matching assignment.



Sarlin et al., "SuperGlue: Learning Feature Matching with Graph Neural Networks," CVPR 2020 ([OpenAccess](#)).

- ▶ Trained to “meta-learn” classification solutions for tabular datasets; pretraining uses millions of **synthetic** datasets, never real data.
- ▶ A new dataset is handled by **in-context learning**: the training rows go in as context, and predictions come out in one forward pass, in under a second.
- ▶ Zero-shot predictions for new, small tabular datasets.
- ▶ TabPFN v2 (published in *Nature*, 2025) extends this to regression, missing values, and larger tables.
- ▶ Applications: Science data analysis, health records, finance.

TabPFN: Architecture and Capabilities



Hollmann et al., "TabPFN: A Transformer That Solves Small Tabular Classification Problems in a Second," ICLR 2023

([arXiv:2207.01848](https://arxiv.org/abs/2207.01848))

- Learns a prior over tabular tasks by pretraining on large synthetic task distributions.
- Performs in-context learning on a new dataset in one forward pass.
- Two-way attention captures dependencies across both rows and features.

- ▶ Model of brain-response dynamics.
- ▶ Frozen encoders for video/audio/text + transformer integration backbone.
- ▶ Supports zero-shot transfer to unseen subjects/tasks at population level.

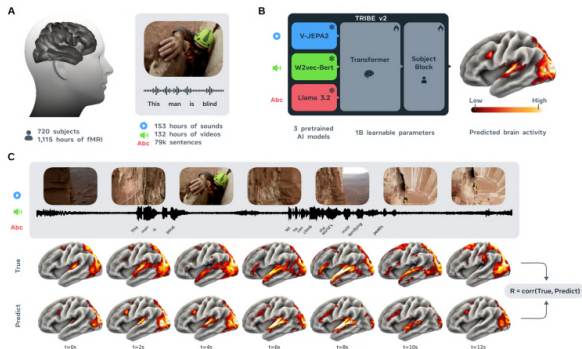


Figure from Meta AI's TRIBE v2 announcement: data → pretrained embeddings → predicted cortical response vs. measured mean response

TRIBE v2: Encoding Performance (Results)

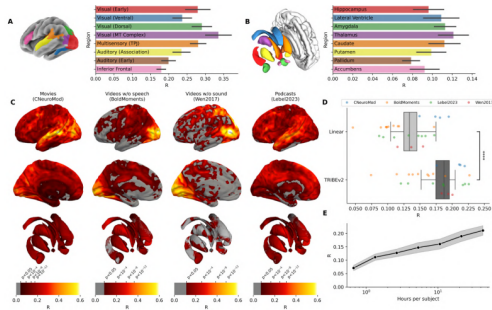
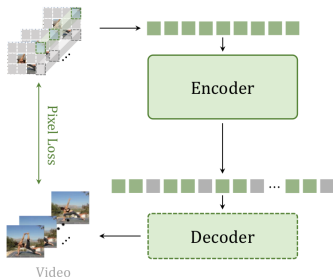


Figure from Meta AI's TRIBE v2 announcement:
Cortical/subcortical encoding scores; vs. linear baseline;
scaling with training data.

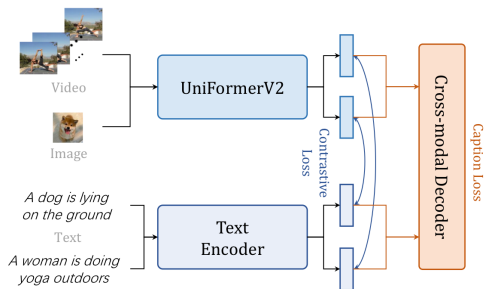
- **Accurate whole-brain prediction:** TRIBE v2 accurately predicts fMRI responses across the entire brain.
- **Cortical encoding scores:** Achieves high encoding scores in key cortical regions.
- **Subcortical encoding performance:** Robust encoding in 8 subcortical regions.
- **Per-subject overview:** Individual subject encoding scores, with dots for TRIBE v2.
- **Scaling laws:** Demonstrates average encoding scores on the Algonauts dataset improve as more training data hours per participant are used.

- ▶ Large-scale pre-trained video foundation model (InternVideo family; InternVideo2, Wang et al., 2024).
- ▶ Developed to generalize across diverse video understanding tasks by learning rich spatio-temporal features adaptable to unseen video domains.
- ▶ Capabilities: Recognition, localization, captioning, segmentation, search, QA, and multi-modal video reasoning.
- ▶ Complements last lecture's V-JEPA 2: same goal of general video representations, but built from masked reconstruction plus contrastive alignment rather than latent prediction.

- Stage 1: Spatiotemporal perception (masked reconstruction)
- Stage 2: Semantic alignment (video-text/audio contrastive learning)
- Stage 3: World-modeling style reasoning (next-token objectives)



(a) Masked Video Encoder



(b) Multimodal Video Encoder

Figure from the InternVideo2 paper: architecture overview.

- ▶ **Homogenization risk:** when many applications share one backbone, they inherit the same failure modes and biases (Bommasani et al.).
- ▶ **Data provenance:** web-scale corpora raise licensing, privacy, and bias questions; curation (DINOv2's LVD-142M) and licensed collection (SA-1B) are partial answers, not full solutions.
- ▶ **Evaluation:** zero-shot numbers depend on prompt engineering, and at web scale it is hard to rule out that benchmark images appeared in training data.
- ▶ **Compute concentration:** pretraining budgets limit who can build these models; most practitioners work at the **adaptation** layer (probes, LoRA, prompting).
- ▶ **Task coverage:** strong global features do not guarantee strong *dense* features (segmentation, depth); DINOv3's gram anchoring targets exactly this gap.
- ▶ **Source bias:** most of today's headline vision foundation models come from a handful of industry labs; treat their benchmark selections critically.

- ▶ Foundation models are reshaping AI across all domains (vision, language, tabular, medicine, video).
- ▶ Key trends: Scalability, data diversity, task generalization, multi-modal grounding.
- ▶ Challenges: Compute costs, ethical issues, responsible deployment, domain adaptation.
- ▶ **Discussion:**
 - How will foundation models change your research?
 - What domains remain out-of-reach for today's foundation models?

- ▶ **Hugging Face transformers:** CLIP (openai/clip-vit-base-patch32, zero-shot classification pipeline), SAM (facebook/sam-vit-huge), DINOv2 (facebook/dinov2-base), Depth Anything V2, and Grounding DINO (IDEA-Research/grounding-dino-tiny) all load in a few lines.
- ▶ **Official repos:** [facebookresearch/segment-anything](#), [facebookresearch/sam2](#), [IDEA-Research/GroundingDINO](#), [PriorLabs/TabPFN](#).
- ▶ **Suggested exercise:** build an open-vocabulary segmentation pipeline: Grounding DINO turns a text prompt into boxes, SAM turns the boxes into masks. Compare against a DINOv2 encoder with a linear head on the same images.

1. Bommasani et al., “On the Opportunities and Risks of Foundation Models.” arXiv:2108.07258.
2. Radford et al., “Learning Transferable Visual Models From Natural Language Supervision (CLIP),” ICML 2021. arXiv:2103.00020.
3. Oquab et al., “DINOv2: Learning Robust Visual Features without Supervision,” TMLR 2024. arXiv:2304.07193.
4. Siméoni et al., “DINOv3,” 2025. arXiv:2508.10104.
5. Liu et al., “Grounding DINO: Marrying DINO with Grounded Pre-Training for Open-Set Object Detection,” ECCV 2024. arXiv:2303.05499.
6. Kirillov et al., “Segment Anything,” ICCV 2023. arXiv:2304.02643.
7. Ravi et al., “SAM 2: Segment Anything in Images and Videos,” ICLR 2025. arXiv:2408.00714.
8. Meta AI, “SAM 3: Segment Anything with Concepts,” 2025. arXiv:2511.16719 (SAM 3.1 update: Meta AI blog).
9. Yang et al., “Depth Anything: Unleashing the Power of Large-Scale Unlabeled Data,” CVPR 2024. arXiv:2401.10891.
10. Yang et al., “Depth Anything V2,” NeurIPS 2024. arXiv:2406.09414.
11. Sarlin et al., “SuperGlue: Learning Feature Matching with Graph Neural Networks,” CVPR 2020.
12. Hollmann et al., “TabPFN: A Transformer That Solves Small Tabular Classification Problems in a Second,” ICLR 2023. arXiv:2207.01848.

13. Hollmann et al., “Accurate Predictions on Small Data with a Tabular Foundation Model,” *Nature*, 2025.
14. Prior Labs, TabPFN resources. <https://github.com/PriorLabs/TabPFN>.
15. Meta AI, “Introducing TRIBE v2: A Predictive Foundation Model Trained to Understand How the Human Brain Processes Complex Stimuli.”
16. Wang et al., “InternVideo2: Scaling Foundation Models for Multimodal Video Understanding,” ECCV 2024. arXiv:2403.15377.
17. Hu et al., “LoRA: Low-Rank Adaptation of Large Language Models,” ICLR 2022. arXiv:2106.09685.
18. Rombach et al., “High-Resolution Image Synthesis with Latent Diffusion Models,” CVPR 2022.
19. Alayrac et al., “Flamingo: a Visual Language Model for Few-Shot Learning,” NeurIPS 2022.
20. Liu et al., “Visual Instruction Tuning (LLaVA),” NeurIPS 2023.
21. Ramesh et al., “Zero-Shot Text-to-Image Generation (DALL-E),” ICML 2021.
22. Schuhmann et al., “LAION-5B: An Open Large-Scale Dataset for Training Next Generation Image-Text Models,” NeurIPS 2022. OpenCLIP: https://github.com/mlfoundations/open_clip.
23. Stanford CS231N/CS231A and BIODS 271/276 lecture resources (as listed in report “Works cited”). <https://web.stanford.edu/class/biods276/lectures/lecture1.pdf>